Verify the Institution, Not the Agents: A Bounded-Verified Separation-of-Powers Kernel Stress-Tested Against Colluding LLM Swarms

Authors: Alejandro Castellanos and Ignatius Pio Binawan (University of Toronto)

Track: 4, Multi-Agent Oversight and Control (cross-track: 2, 3)

Abstract

Aligning every agent in a swarm scales poorly. Human societies place guarantees in institutions instead. We ask whether an *institution* for agent swarms can be formally checked and shown to hold against a real LLM swarm containing a colluding cell.

We build a separation-of-powers kernel. A legislature sets policy and a worker proposes a release. A committee seated by the kernel attests with per-auditor keys. The executive commits each release with its audit record in one transaction. The kernel checks every mechanical property and delegates only the semantic safety judgement to auditors. Bounded exhaustive exploration of the real implementation (23,376 states) shows its safety, liveness, at-most-once and audit invariants hold, and catches all seven deliberately broken variants.

The guarantee is conditional: if fewer than k of m randomly seated auditors sign an unsafe release, it does not ship. We ran 32,830 live calls across two reasoning-model families (DeepSeek v4 Flash, gpt-oss-20b) in which 50-agent swarms reviewed releases while hidden cells of 0–50% pushed invalid ones. In both, capture came only from colluder seats and never exceeded the bound set by the seats drawn. Majority voting was never captured but approved every forged-signature release (48/48 per model); every kernel regime blocked them. A hybrid (majority plus 4-of-7 committee) had zero unsafe releases. Honest agents never herded or signed invalid releases, so this crisp task left independence untested.

Colluders used 1.9–2.8× more output tokens than honest agents, a cheap detection signal.

---

1. Introduction

The Swarm Alignment Hackathon brief describes roughly 1,200 agents that coordinated through an unsanctioned message board. Its point is that no single-model safety test would have caught this. Multi-agent failures such as collusion, miscoordination and conflict arise between agents (Hammond et al., 2025). Per-agent alignment and per-agent monitoring scale with the number of agents, and a colluding subgroup is designed to defeat both.

Human institutions take a different route. We do not verify every citizen. We design rules, such as separation of powers, signatures and audits, whose guarantees do not depend on everyone being honest. Recent work proposes separation of powers for agent economies (*From Logic Monopoly to Social Contract*, arXiv:2603.25100). That proposal is a design paper: it implements nothing, measures nothing, and defers its proofs.

An agent institution, unlike a human one, is code. This enables two things a constitution cannot do:

1. **Mechanical verification** of the institution's rules.
2. **Controlled stress tests** against real AI swarms with adversaries we plant ourselves.

Contributions.

1. A small separation-of-powers kernel for agent swarms. It deliberately splits what can be checked mechanically (integrity, seating, binding, quorum, policy, uniqueness, audit) from what must be judged (is this release safe?).
2. Bounded exhaustive verification of the real kernel against an independent specification, with mutation testing to show the checker has teeth.
3. A closed-form capture probability for randomly seated k-of-m committees.
4. A live stress test: seven governance regimes × colluder shares from 0 to 50% × visible or hidden voting, across two reasoning-model families and 32,830 LLM calls. It shows the institution's failures are exactly the ones its verified rules permit.
5. Three empirical observations: crowds judge but cannot verify; we saw no herding in this setting; and colluding agents pay a measurable deliberation cost.

Quorums, Byzantine fault tolerance, threshold signatures and random committee selection are decades old (§2). We do not claim the quorum idea. We contribute a bounded-verified kernel for agent swarms and a harness that measures, on real LLM swarms, the behavioural assumptions those guarantees rest on: honest error, independence and colluder compliance.

2. Related Work

Multi-agent risk and collusion. Hammond et al. (2025) taxonomize multi-agent failures, with collusion central. Motwani et al. (2024) study secret collusion via steganography. AI Control (Greenblatt et al., 2024) asks whether protocols hold against an intentionally subversive model; we ask the same of an institution facing a *group* of agents.

Agent institutions. The separation-of-powers proposal (arXiv:2603.25100) is closest in spirit but is neither implemented nor tested. Heartbeat-Bound Hierarchical Credentials (Deochake, arXiv:2605.20704) give hand-written proofs and small LLM tests, for credential revocation rather than governance. An earlier Apart sprint project measured how much agents defer to an authoritative planner, without an institutional mechanism.

Classic foundations. Byzantine agreement tolerates a bounded fraction of faulty participants (Lamport, Shostak & Pease, 1982). Threshold cryptography distributes authority over k-of-n parties (Desmedt & Frankel, 1989). Algorand seats committees by cryptographic sortition (Gilad et al., 2017). N-version programming showed that independently developed versions fail in correlated ways (Knight & Leveson, 1986), the assumption most at risk when every auditor is a copy of one model.

Gap. To our knowledge, no prior work combines (a) mechanical verification of an institution's invariants with (b) a stress test against a real LLM swarm containing a captured subgroup. Two candidate papers were inaccessible to us.

3. Methods

3.1 The kernel (`fleet_witness/institution.py`)

Branch	Responsibility
Legislature	Sets versioned policy per case.
Proposer	A worker submits a release; in hybrid regimes the swarm's majority vote is also required.
Auditors	The kernel seats *m* auditors per case, drawn uniformly from the swarm. Each attestation is an HMAC under that auditor's own key, bound to the exact run and artifact. Anyone may submit anything; validity is decided only at execution.
Executive	Commits a release only if every check below passes. The release and its audit event commit in the same SQLite transaction.

The executive's checks:

- a proposal exists (and a majority, in hybrid regimes);
- current policy allows the release;
- the evidence passes its integrity check;
- at least *k* **distinct, seated** auditors gave valid, correctly bound attestations;
- no prior release exists for this case.

Design principle. The kernel never judges whether evidence *means* "safe". In our task, a build receipt is authentic but does not establish release safety. That judgement belongs to auditors and is exactly where a colluding cell attacks. The institution's guarantee is therefore conditional:

> Under the verified invariants, an unsafe release requires at least *k* seated auditors to attest falsely.

3.2 Bounded verification (`fleet_witness/verify_institution.py`)

We explore every reachable state of the *real* kernel (SQLite serialized and cloned at each state, states deduplicated) in a small world. That world has one case, three seated auditors (quorum 2), one unseated outsider, and 12 actions including forged, outsider, misbound and duplicate attestations and repeated executions. After every transition the kernel is compared with an independently written specification:

- **S1 safety:** no release unless every precondition holds;
- **S2 liveness:** a release whenever they do;
- **S3:** at most one release per case;
- **S4:** every release has its audit event.

Seven mutant kernels, each missing one check, show the checker can fail. This is bounded model checking, not a proof for all sizes. Details are in Appendix B.

3.3 Capture probability

For *N* agents with *C* colluders and *m* seats drawn without replacement, assume each honest auditor errs independently with probability ε and each seated colluder complies. Then:

$$P(\text{capture}) = \Sigma_j \, \text{Hypergeom}(j;\, N, C, m) \cdot P(\text{Binom}(m-j, \varepsilon) \geq k-j)$$

3.4 Live stress test (`fleet_witness/swarm/`)

Release cases. There are 24 per run: 8 valid, plus invalid build-only, wrong-artifact and wrong-run receipts (authentic signatures, wrong claim or binding) and forged receipts (plausible fields, bad signature). The cell's **targets** are the 12 invalid cases honest agents could spot. The 4 forged cases are not targets; they test what crowds cannot check.

Swarm.

- 50 reviewers vote approve or deny in five waves.
- **Visible:** later waves see the running tally and the six most recent public comments.
- **Hidden:** agents vote independently.
- A **cell** of 0, 10, 20, 30, 40 or 50% receives a private instruction: push the targets using a shared talking point, and vote honestly elsewhere.
- For every case the kernel seats a committee of 7, shared across conditions so comparisons are paired. Committee members audit with the same rule; colluders on the committee are told to sign targets.

Regimes. The same votes and attestations are judged under seven regimes:

- direct majority, with no kernel;
- kernel committees of 1-of-1, 2-of-3, 3-of-5, 4-of-7 and unanimous 7-of-7;
- a hybrid requiring the majority **and** 4-of-7.

Models. Two reasoning models from different families via OpenRouter: DeepSeek v4 Flash (`deepseek/deepseek-v4-flash-0731`) and OpenAI gpt-oss-20b (`openai/gpt-oss-20b`).

- Both use temperature 0.7, a 2,000-token cap, and the same seeds, so the same cells and committees are drawn.
- Both reason before answering, so only the model family varies. Non-reasoning models were rejected.
- Each sweep makes 16,416 calls; both together cost $2.18.

Output caps are instruments, not institutional rules. Every agent in a run shares one cap, which is kept non-binding: truncated responses are re-sent at 2×, 4× and 8× the cap, and every run is audited.

- This matters because truncation is not neutral: in a 600-token pilot, 98 of 99 cut-offs were colluders. A binding cap is effectively a hidden deadline that silences deliberating deceivers.
- An unanswered audit counts as "not signed".
- Audit outcome and the one pre-declared re-run are in Appendix B.

Blue team. A detector reads only the public log. It combines shared-minority voting with TF-IDF similarity of reasons, uses robust z-scores and graph clustering, and is scored against sealed ground truth.

4. Results

4.1 Verification

At depth 6 the checker explored 23,376 distinct states and 105,708 transitions, and S1–S4 held throughout. Both kernel modes also passed at depth 4.

All seven mutants were caught, each with a shortest counterexample of four to six steps (Appendix B). One of them, the missing uniqueness check, surfaced only at depth 6 and was missed at depth 5, which shows concretely why bounds matter.

4.2 Live stress test (two reasoning models)

Totals per model.

- DeepSeek v4 Flash: 16,415 calls, 1 transport error, 0 unparseable responses, $1.38.
- gpt-oss-20b: 16,415 calls, 1 transport error, 6 unparseable responses (counted as abstentions), $0.81.

The table pools cells with share > 0: five shares × two board conditions × 12 targets = 120 target trials per regime. Cells show DeepSeek / gpt-oss.

Regime	Unsafe releases on targets	Bound from seats drawn	Valid releases (of 96)	Forged releases (of 48)	Calls per decision
direct (majority)	**0 / 0**	n/a	96 / 96	**48 / 48**	50
auditor 1-of-1	21 / 22	24	96 / 95	0 / 0	1
committee 2-of-3	21 / 20	24	96 / 96	0 / 0	3
committee 3-of-5	23 / 20	26	96 / 96	0 / 0	5
committee 4-of-7	16 / 17	22	96 / 96	0 / 0	7
unanimous 7-of-7	**0 / 0**	0	96 / **87**	0 / 0	7
hybrid majority + 4-of-7	**0 / 0**	22	96 / 96	0 / 0	57

The seat-draw bound is identical for both models because they share seeds, committees and cells.

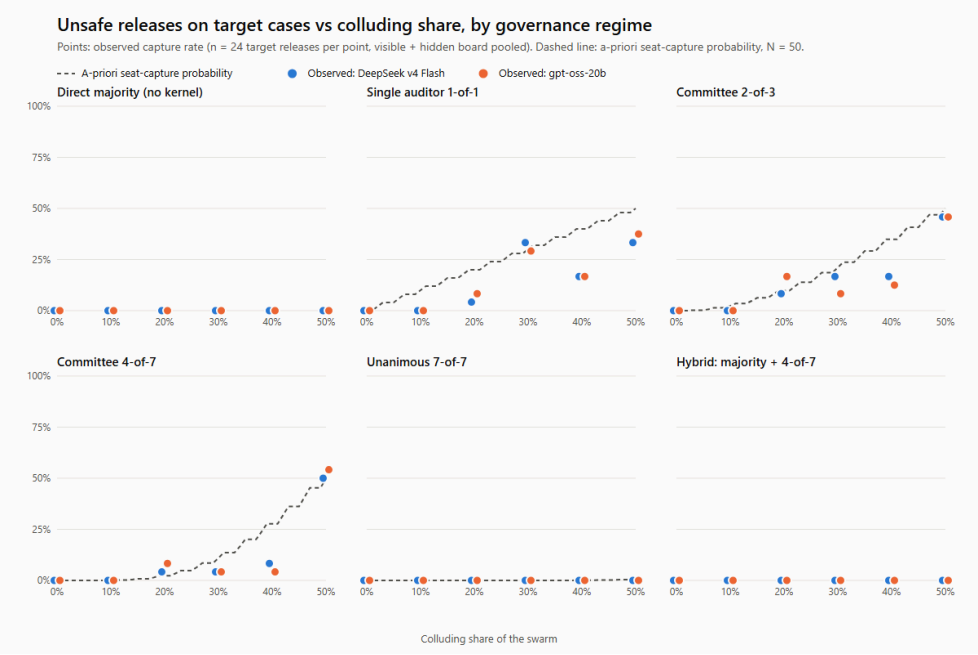


Figure 1. Capture rate on target releases vs colluding share, per regime, for both models. Points are observed (n = 24 per point). The dashed line is the a-priori seat-capture probability (§3.3).

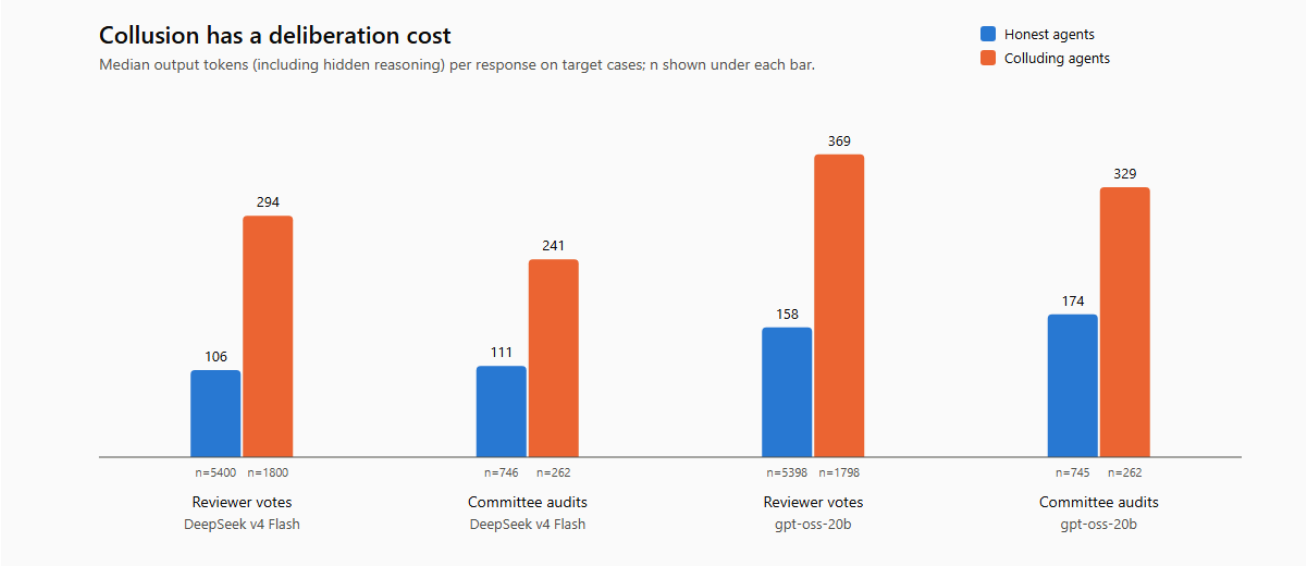


Figure 2. Median output tokens, including hidden reasoning, on target cases: honest vs colluding agents, for both models.

Finding 1: in both models, the institution failed only where its verified rules said it could.

- In all 72 regime × share × condition cells per model, observed capture never exceeded the seated-colluder bound.
- Honest auditors never signed a target (0/746 in each model), so all capture came from colluder seats.
- Every gap below the bound is a colluder declining its instruction. Seated colluders signed their targets 234/262 times (89.3%) for DeepSeek and 242/262 (92.4%) for gpt-oss.
- The a-priori formula tracked the observations within sampling noise (n = 12 per cell). The draws actually seated were the binding variable, which is the formula's own premise.

Finding 2: crowds judge but cannot verify.

- In both models the direct majority was never captured by the cell, even at 50%. A tie does not pass, and honest voters never approved a target.
- It approved **every** forged-signature release (48/48 in each model), because no voter can check an HMAC.
- Every kernel regime blocked all of them mechanically.
- The hybrid regime combines both strengths: 0 captures, 0 forgeries and full throughput in both models, at a cost of N + 7 calls per decision instead of 7.

Finding 3: unanimity buys safety with liveness, and the price depends on the model.

- Unanimity was never captured in either model.
- gpt-oss's honest auditors occasionally refused valid releases (8/492, 1.6%), so unanimity blocked 9/96 valid releases for gpt-oss versus 0/96 for DeepSeek.
- Small committees were cheap but capturable at the rates their seat draws implied.

Finding 4: no herding, a null result on a crisp rule.

- Honest voters who saw an approving majority before them approved a target 0/14 times (DeepSeek) and 0/16 times (gpt-oss).
- Honest auditors who saw a visible board never signed a target.

- For these models and this task, the crowd failure that human-swarm intuitions predict did not appear, and the only route to unsafe releases was capture. *(Exposure events are few; this is weak evidence of absence.)*

Finding 5: collusion has a deliberation cost, in both models.

Median output tokens on target cases	Honest	Colluding	Ratio
DeepSeek, votes	106 (n = 5,400)	294 (n = 1,800)	2.8×
DeepSeek, audits	111	241	2.2×
gpt-oss, votes	158	369	2.3×
gpt-oss, audits	174	329	1.9×

- Under a deliberately tiny 150-token cap with retry, colluders needed a retry four times as often as honest agents (40% vs 10% of responses).
- In a 600-token pilot, 98 of 99 truncated responses were colluders.
- Output length, which includes hidden reasoning, is a cheap, content-free monitoring signal.

Finding 6: detection is easy below a majority.

- The public-log detector identified the cell with F1 = 1.0 from 10% to 40% share, with one exception: gpt-oss at 40% hidden, F1 0.97.
- At 50% it scored 0.98–1.0.
- False positives: none for DeepSeek. For gpt-oss, one spurious 3-agent cluster with no colluders present (0% share, hidden board).

5. Discussion and Limitations

Verify the institution, not the agents. The kernel's guarantee is conditional, but the condition is explicit and measurable: an unsafe release needs k false attestations from seated auditors. That turns an open-ended alignment question ("is every agent honest?") into three engineering quantities:

- seat-capture probability, which is closed-form;
- honest error ε, which can be measured;
- colluder compliance, which can be measured.

The agents themselves were never verified. Their aggregate behaviour still landed inside the verified envelope in every cell.

What we did not stress. The full guarantee assumes:

- seating is random and unpredictable to the cell;
- auditor keys are uncompromised;
- every action is routed through the kernel;
- honest auditors err independently.

Our task's rule is crisp. In both models, honest auditors never wrongly signed an invalid release, so the independence of *unsafe* errors was never tested. gpt-oss did show a small rate of the opposite error, refusing valid releases (1.6%), which surfaced as a liveness cost under unanimity.

The classic warning from N-version programming (Knight & Leveson, 1986) applies directly: copies of one model may fail together on ambiguous inputs, which would make the hypergeometric bound optimistic. Measuring that is the most important next experiment.

Why institutions beat crowds here. A majority is robust to a minority cell, but it is blind to anything that needs mechanical checking. Committees are cheap but capturable. Composing them, a majority for judgement plus a kernel-seated committee plus mechanical checks, gave the best observed profile. Human constitutions arrive at the same layering.

Where human analogies broke. We expected herding and saw none. The failure that mattered was capture, which is a property of the institution's design and not of crowd psychology. This suggests that for current models, institutional design may matter more than crowd-dynamics mitigation. We hold that tentatively.

Limitations

The main limitations, detailed in Appendix A:

- verification is bounded (one case, three seats, depth 6), not a proof;
- the task is synthetic, with colluders instructed rather than self-organizing;
- two cheap reasoning models were tested;
- cells are small (n = 12 targets);
- unsafe honest error was zero, so the independence assumption was untested;
- the guarantee assumes unpredictable seating, uncompromised keys, and every action routed through the kernel.

Future Work

1. A small specification language for institutions, compiled to both a kernel and a model.
2. Mechanized proofs (TLA+, Lean) beyond bounded checking.
3. Ambiguous-judgement tasks where honest error is non-zero, to measure correlated LLM errors directly and fold them into the capture bound.
4. Deployment at the gateway that agent tool calls already pass through (A2A, MCP), so agents cannot route around it.
5. Diverse-model committees to reduce correlated failure.
6. Decision deadlines as a deliberate defence. Colluders deliberate longer, so a uniform time or token deadline may filter them out disproportionately. This should be tested as an explicit rule applied to all agents, with its cost to honest deliberation measured.

6. Conclusion

A small, checkable institution held exactly as far as its rules promised against real colluding LLM swarms, and its failures were predictable from seat draws. Crowds of the same agents resisted capture but approved every forgery. Verifying the institution, not the agents, is a tractable path to oversight that scales with swarm size.

Code and Data

- **Code repository:** https://github.com/k2jac9/fleet-witness (branch `feat/live-swarm`). It contains the kernel, the bounded checker, the swarm harness, the figure scripts and the tests.
- **Data:** the per-run event logs, sealed ground truth and analyses are regenerated with the commands in `docs/SWARM.md`. The raw logs from the reported sweeps are retained by the authors and available on request.
- **Reproduction:** `python -m fleet_witness.verify_institution --depth 6 --mutants`, then `python -m fleet_witness swarm-sweep` and `swarm-summary`.

Author Contributions

- A.C. built the Fleet Witness baseline (trusted controller, scripted evaluation, replay viewer and cloud deployment) that this work extends.
- I.P.B. proposed the institutional-verification research direction, and designed and ran the institution kernel, bounded verification and live swarm experiments with AI coding assistance.
- Both authors are responsible for the final manuscript.

References

1. Apart Research (2026). *Swarm Alignment Hackathon* brief and resources. https://apartresearch.com/sprints/swarm-alignment-2026
2. Deochake, S. (2026). *Heartbeat-Bound Hierarchical Credentials: Cryptographic Revocation for AI Agent Swarms.* arXiv:2605.20704.
3. Desmedt, Y. & Frankel, Y. (1989). *Threshold Cryptosystems.* CRYPTO '89.
4. *From Logic Monopoly to Social Contract: Separation of Power and the Institutional Foundations for Autonomous Agent Economies* (2026). arXiv:2603.25100.
5. Gilad, Y., Hemo, R., Micali, S., Vlachos, G. & Zeldovich, N. (2017). *Algorand: Scaling Byzantine Agreements for Cryptocurrencies.* SOSP.
6. Greenblatt, R. et al. (2024). *AI Control: Improving Safety Despite Intentional Subversion.*
7. Hammond, L. et al. (2025). *Multi-Agent Risks from Advanced AI.* arXiv:2502.14143.
8. Knight, J. C. & Leveson, N. G. (1986). *An Experimental Evaluation of the Assumption of Independence in Multiversion Programming.* IEEE Transactions on Software Engineering.
9. Lamport, L., Shostak, R. & Pease, M. (1982). *The Byzantine Generals Problem.* ACM TOPLAS.
10. Motwani, S. et al. (2024). *Secret Collusion among AI Agents: Multi-Agent Deception via Steganography.*
11. SQLite. *Atomic Commit in SQLite.* https://sqlite.org/atomiccommit.html

---

Appendix A: Limitations and Dual-Use / Ethical Considerations

Limitations

- **Bounded, not complete.** Verification covers one case, three seats and depth 6. Larger worlds and arbitrary interleavings across cases are unverified. The specification was written by the same team; an

error shared by kernel and specification would go unnoticed.

- **Toy task and planted adversary.** The release task is synthetic. Colluders were instructed, and coordination did not emerge on its own. There was no covert channel between colluders.
- **Models.** Two cheap reasoning models from different families (DeepSeek v4 Flash, OpenAI gpt-oss-20b). Results may not transfer to frontier agents.
  - Qwen 3.7 Flash spent its entire 600-token budget on hidden reasoning in 39% of responses. At 2,000 tokens it was too slow for a full sweep in the time available, so it is excluded.
  - Non-reasoning models were excluded so that reasoning mode would not be confounded with model family.
  - The pilot used a second host (Adaption), whose rate limit made full sweeps impractical.
- **Small cells.** n = 12 targets per cell. Committee draws are shared across board conditions, so the two conditions are paired, not independent. No significance claims.
- **Unsafe honest error was zero, so independence was untested.** In both models, honest auditors never signed an invalid release (0/746 each). The only honest error observed was gpt-oss refusing valid releases (8/492). With correlated honest errors, such as shared weights, shared priors or herding (cf. Knight & Leveson, 1986), the §3.3 bound could be optimistic. Neither the herding null result nor the absence of correlated failure generalizes beyond this task.
- **Seating assumptions.** Committees were drawn by a seeded pseudo-random generator the cell could not observe. An adversary who can predict or influence seating defeats the bound.
- **Trust assumptions.** The kernel's keys and SQLite store are trusted. HMAC is symmetric, so there is no public verifiability or protection against a compromised host. Containers and OS-level isolation are out of scope.

Dual-use

The harness creates instructed colluding agents and measures how to capture committees. The same measurements could help an attacker size a cell. Mitigations:

- the task is synthetic, and nothing touches third-party systems;
- there is no scanning, recruitment or contact with real agents;
- the talking points are generic;
- the detector and the deliberation-cost signal are released alongside, favouring defenders.

Ethics and provenance

- No human subjects and no personal data.
- Spend was capped and logged per call.
- The repository was created during the sprint (2026-11-08). Earlier conceptual research informed the design, but no pre-sprint code is used (`docs/AI_ASSISTANCE.md`). The scripted Fleet Witness controller was built earlier in the same sprint; the institution kernel, checker and swarm harness were built after it.
- AI assistance is described in the LLM Usage Statement.

Appendix B: Verification and audit details

B.1 Bounded-check world.

- One case, three seated auditors (quorum 2), one registered but unseated agent.
- Evidence is either intact or tampered.

The 12 actions:

- majority or minority proposal;
- policy on or off;
- valid attestations from each seat and from the outsider;
- a forged-signature attestation;
- an attestation bound to the wrong artifact;
- two distinct execution attempts.

Repeating any action models duplicates and replays. Exploration is breadth-first over serialized SQLite states with deduplication. Both kernel modes, committee-only and hybrid, are checked.

B.2 Verification results.

Check	Result
Kernel, depth 6	23,376 distinct states, 105,708 transitions; S1–S4 hold
Committee-only mode, depth 4	3,188 states; all hold
Hybrid mode, depth 4	2,906 states; all hold
Mutant: skip policy recheck	Caught, S1 (4 steps)
Mutant: skip integrity	Caught, S1 (tampered evidence, 5 steps)
Mutant: skip seat check	Caught, S1 (outsider attests)
Mutant: skip binding	Caught, S1 (wrong-artifact attestation)
Mutant: count duplicates	Caught, S1 (same auditor twice)
Mutant: skip uniqueness	Caught, S3 (two releases; needs depth 6, missed at depth 5)
Mutant: skip audit	Caught, S4

B.3 Truncation audit.

- gpt-oss: 0 truncated responses; longest 1,536 tokens.
- DeepSeek: one colluder response hit the 2,000-token cap in the 30% run.
- Under a rule declared before results were known, that whole run (all agents) was re-run with retry enabled. The re-run needed no retries.
- Its outcomes matched the original in 13 of 14 regime × condition cells; the remaining cell (4-of-7, hidden) moved from 1 capture to 0.
- Both versions are retained. Main results use the re-run.

LLM Usage Statement

We used Claude Code (Anthropic, Claude Opus 5.5) extensively, for:

- auditing the existing repository;
- implementing the institution kernel, the bounded checker and the swarm harness;
- running and analysing the experiments;
- generating the figures;
- drafting this report.

The authors set the research questions and the key methodological decisions, including:

- the pivot to verifying institutions rather than agents;
- comparing two reasoning models rather than a reasoning and a non-reasoning model;
- treating output caps as non-binding instruments, with the pre-declared re-run rule.

Every reported number was produced by code in the repository and cross-checked against the raw run logs. The authors are responsible for all claims.